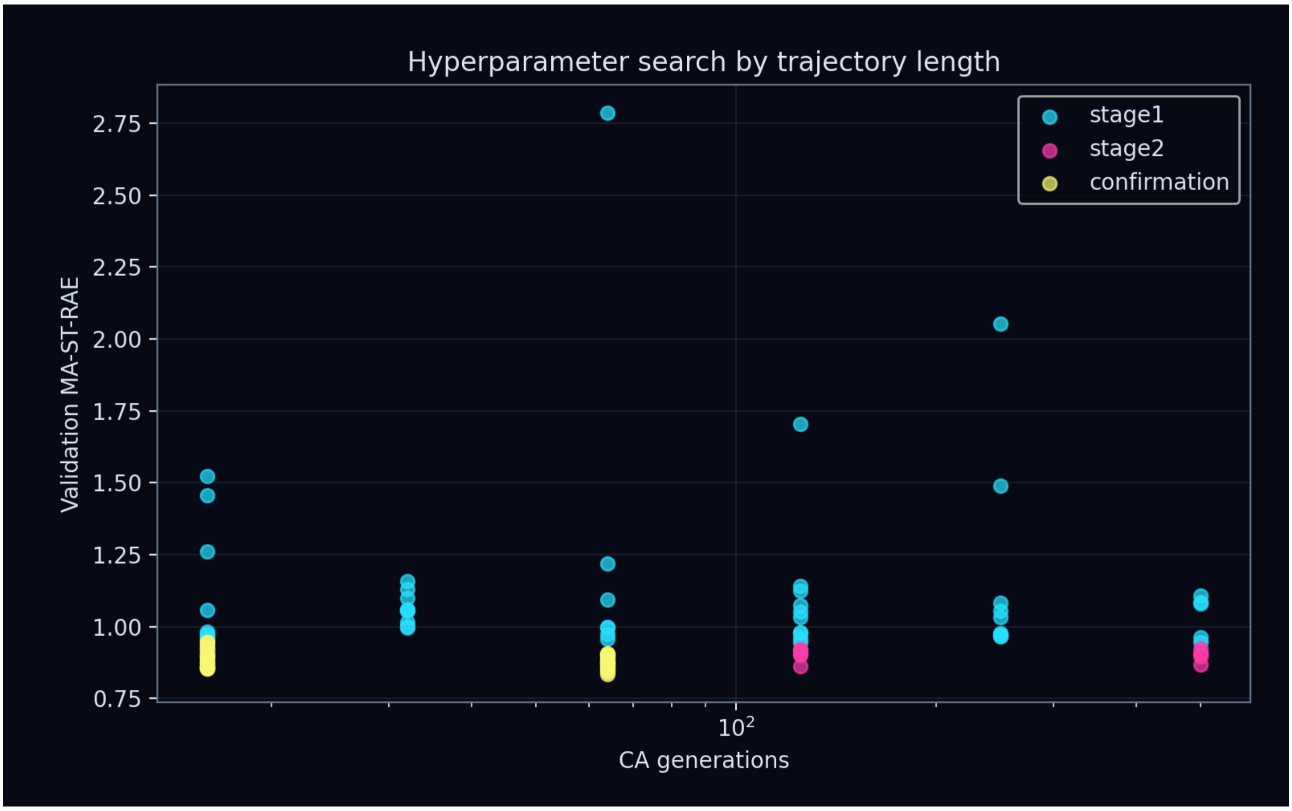
Strange Matter Engine

Production study: kuramoto_sakaguchi

Direct-inhibition pIC50 | grouped validation | blinded set excluded

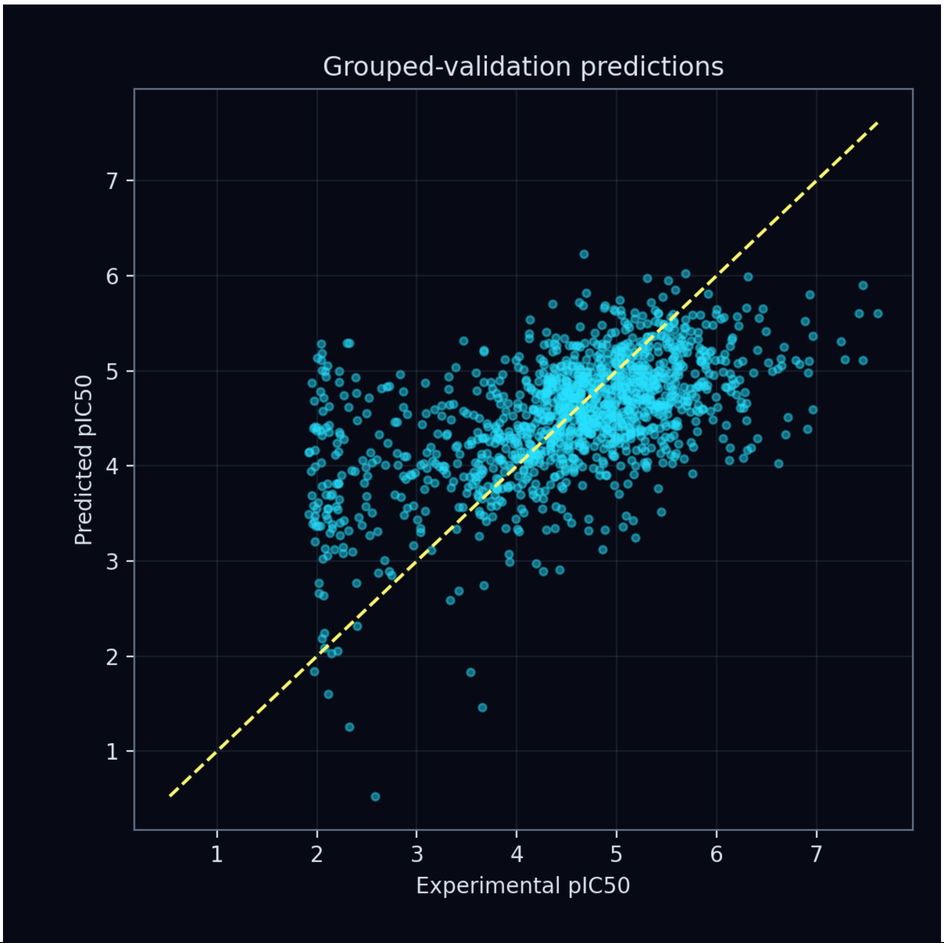
Quantity	Result
Validation MA-ST-RAE	0.8297
Fit MA-ST-RAE	0.8145
Validation RMSE	0.8945 pIC50
Fit RMSE	0.8258 pIC50
Generations	64
Dynamical channels	16
Atom feature profile	local_environment
CA learning rate	0.003
Ridge penalty	1.0
CA L2	1e-06
Gradient clip	1.0
Confirmation mean	0.8754
Confirmation seed SD	0.0221
Bootstrap MA-MAE	0.6584 [0.6245, 0.6921]
Bootstrap MA-R2	0.2290 [0.1849, 0.2732]
Bootstrap MA-Spearman rho	0.4677 [0.4202, 0.5132]
Bootstrap MA-Kendall tau	0.3307 [0.2953, 0.3647]

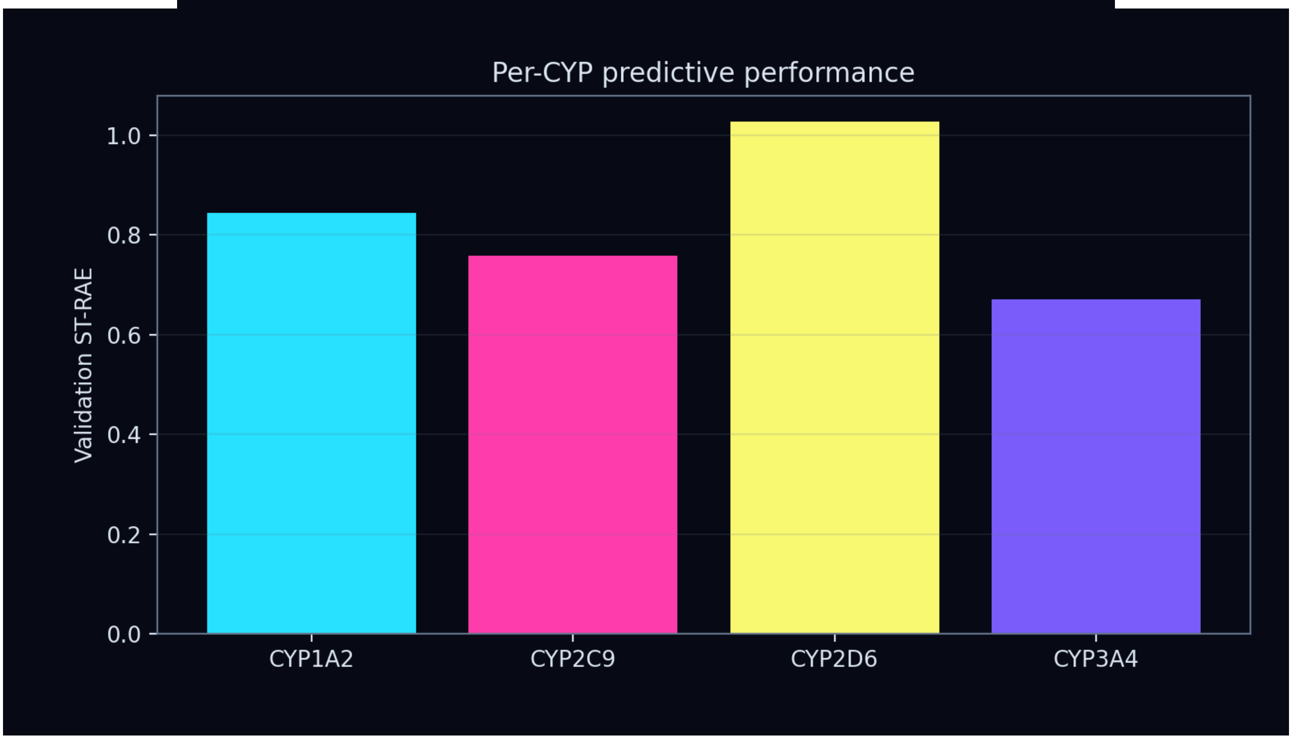
Enhanced CA controls: update scale 0.08, initial-state scale 1.0, support fraction 0.6, bond temperature 0.5. Rule dynamics: A=0.5, B=0.8, C=0.8, D=0.05.


Hyperparameter search by trajectory length
stage1
stage2
confirmation
Validation MA-ST-RAE
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
10^2
CA generations

Predictive validation

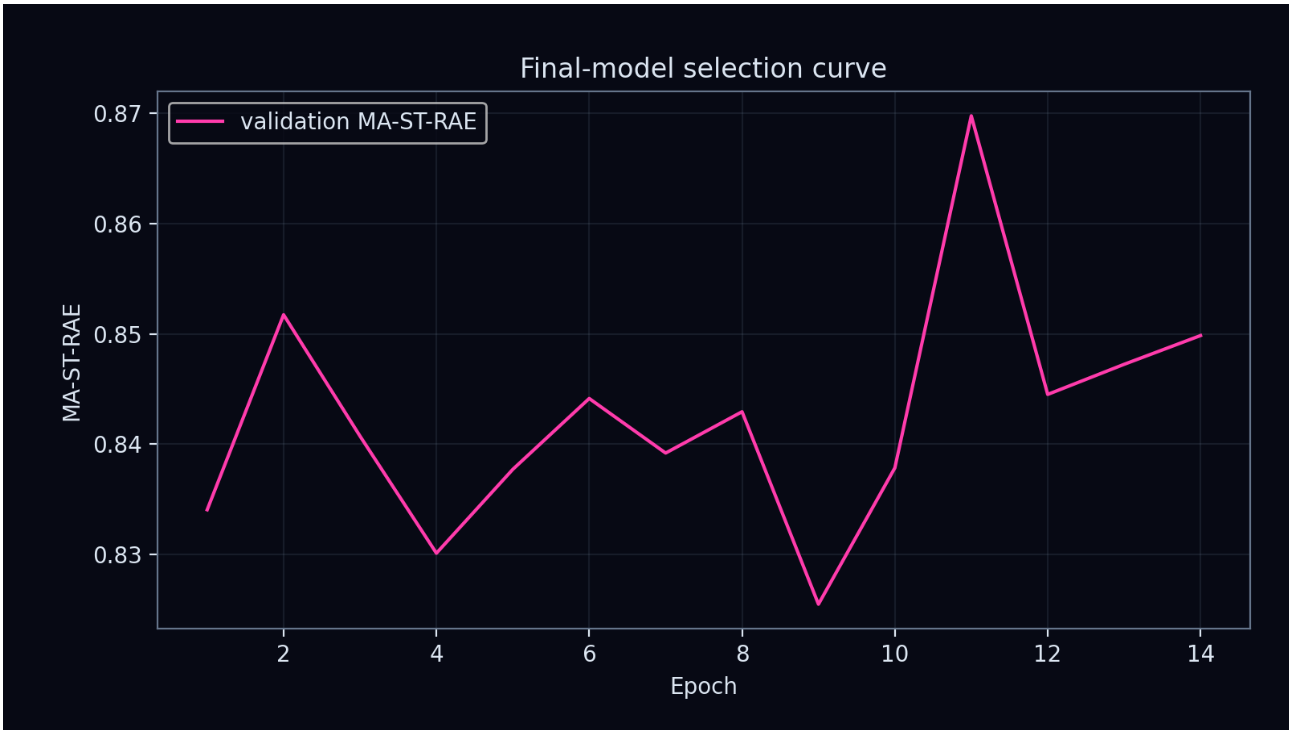
Model selection used grouped-validation MA-ST-RAE, the challenge primary metric. Dynamical-interest scores did not influence promotion.





Optimization

Ridge coefficients and the unpenalized intercept were solved from permitted fitting fingerprints. Query error differentiated through the solve into the graph CA; Adam updated only CA parameters. Bond-conditioned messages, cosine learning-rate decay, and multitime trajectory statistics were shared across all ten rules.

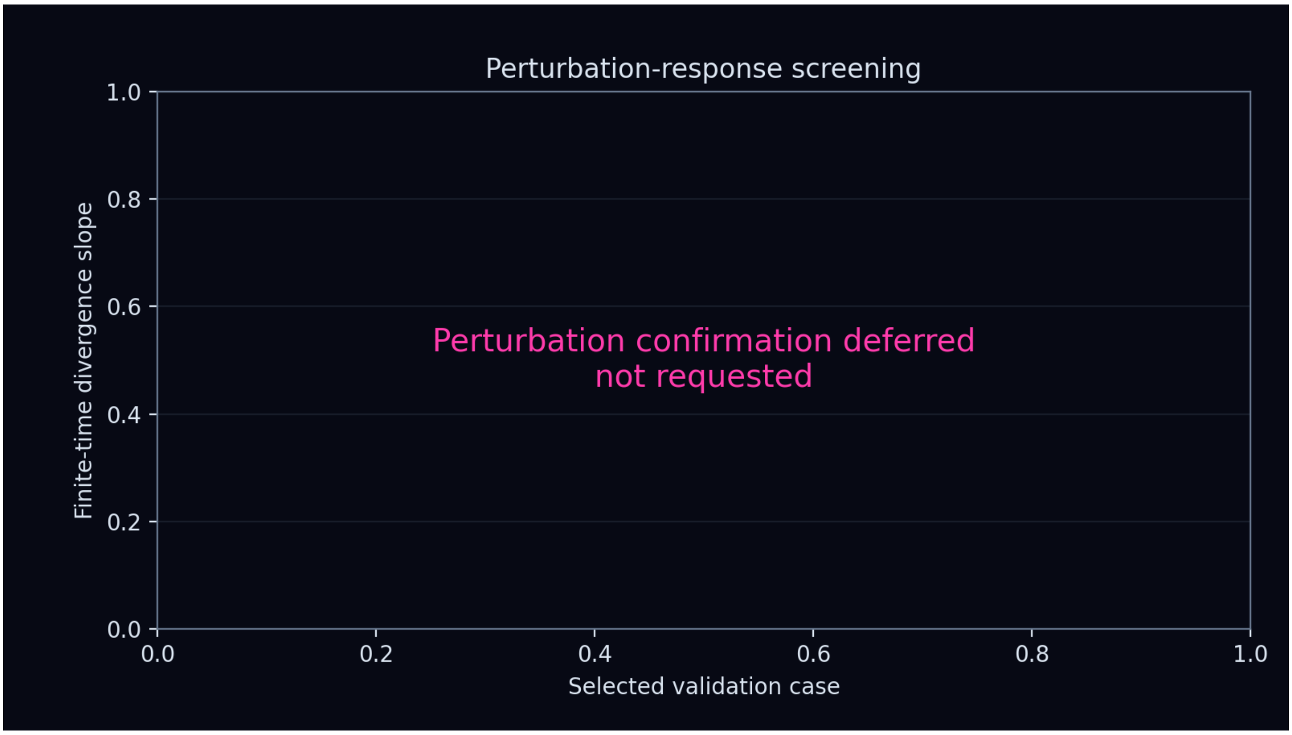


Emergent dynamics

The validation trajectories were screened for convergence, recurrence, persistent motion, spectral complexity, and perturbation sensitivity. Labels ending in candidate require longer, renormalized and seed-repeated confirmation before any attractor or chaos claim.

Permutation and perturbation screening

This figure tests whether small state perturbations separate or contract over the observed finite trajectory. It is placed on a dedicated page to preserve its axes, labels, and complete plotting area.



Scientific limitations

This report describes one transition-rule study under the declared grouped split. Hyperparameter selection and early stopping used labelled training data only. The blinded challenge set was not loaded or predicted. Finite-time positive slopes are screening signals rather than proof of a strange attractor or sustained chaos.